

Reading and interpreting data from pie charts

The same rule run backwards — and the questions a pie chart is not able to answer.

LESSON 164–166

3 × 40 MIN

MATEMATIKA 6, §33

S7 16 INTERPRETING RESULTS

LESSON CLOCK

30'

30'

35'

20'

5

From an angle back to a frequency	30 min
Angles as percentages	30 min
Comparing two pie charts	35 min
What a pie chart cannot tell you	20 min
Homework	5 min

LEARNING OBJECTIVES

- Turn a sector angle back into a frequency when the total is known.
- Express a sector as a percentage of the whole.
- Compare two pie charts drawn from different totals.
- State what a pie chart cannot show, and what extra information is needed.

TEXTBOOK

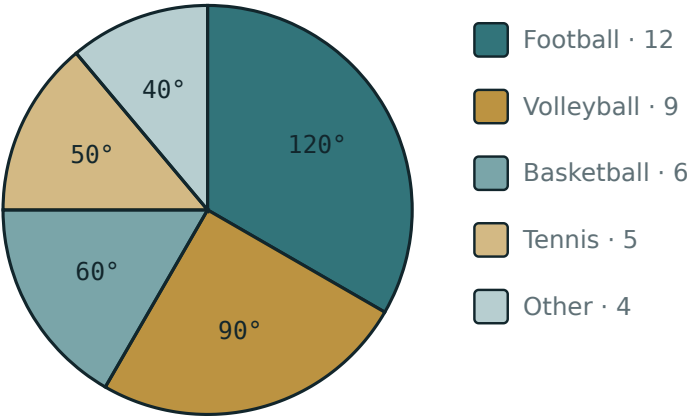
National	pp. 470–479
Cambridge	Stage 7, pp. 178–184
Workbook	Exercise 16.2

01

Explanation

From an angle back to a frequency

Drawing the chart divided 360 by the total. Reading it does the reverse: divide the angle by what one person was worth.



36 pupils · each pupil is 10°

The same chart, now being read rather than drawn

$$\text{frequency} = \frac{\text{angle}}{360^\circ} \times \text{total}$$

TOTAL ASKED	SECTOR	WORKING	FREQUENCY
36	120°	$120 \div 10$	12
60	54°	$54 \div 6$	9
200	90°	$\frac{90}{360} \cdot 200$	50
500	72°	$\frac{72}{360} \cdot 500$	100
240	45°	$\frac{45}{360} \cdot 240$	30

THE TOTAL MUST BE GIVEN SOMEWHERE

A 90° sector is a quarter of *something*. Without the total written on the chart, it is a quarter of an unknown number — and no frequency can be recovered at all.

Angles as percentages

A percentage needs no total: it is the share itself. Divide the angle by 3.6.

$$\text{percentage} = \frac{\text{angle}}{3.6}$$

SECTOR	WORKING	PERCENTAGE	AS A FRACTION
180°	$180 \div 3.6$	50%	$\frac{1}{2}$
144°	$144 \div 3.6$	40%	$\frac{2}{5}$
90°	$90 \div 3.6$	25%	$\frac{1}{4}$
54°	$54 \div 3.6$	15%	$\frac{3}{20}$
36°	$36 \div 3.6$	10%	$\frac{1}{10}$

JUDGE BY EYE, THEN CONFIRM

A sector that looks like a quarter should come out near 25%. If the arithmetic gives 2.5% or 250%, the division went the wrong way round.

Comparing two pie charts

Two charts of the same size can stand for very different numbers of people. The angles compare **shares**; only the totals compare **numbers**.

SCHOOL	PUPILS	FOOTBALL SECTOR	FOOTBALL PUPILS
A	200	90°	50
B	500	90°	125
C	180	120°	60
D	400	90°	100

A LARGER SHARE CAN BE A SMALLER NUMBER

School C gives football a third of its chart and school D only a quarter — yet D has 100 footballers and C has 60. Comparing the angles alone would have reversed the answer.

What a pie chart cannot tell you

QUESTION	CAN THE CHART ANSWER IT?	WHAT IS NEEDED
which group is the largest?	yes	nothing — the biggest sector
what share does a group have?	yes	the angle
how many people is that?	no	the total surveyed
has it changed since last year?	no	a second chart and both totals
what is the mean?	no	the original values, not the shares

THREE SENTENCES MAKE A GOOD INTERPRETATION

The largest group and its share; a comparison between two groups; and one honest limit — usually the sample, its size, or something the chart cannot show.

02 Worked examples

EXAMPLE 1 A pie chart of 60 pupils has a 54° sector for tennis. How many pupils is that?

1 $360 \div 60 = 6^\circ$ for each pupil.

2 $54 \div 6$.

3 = 9 pupils.
Check: $9 \cdot 6 = 54$ ✓

ANSWER

9 pupils

EXAMPLE 2 A survey of 500 people has a 72° sector. Find the number and the percentage.

$$\textcircled{1} \quad \frac{72}{360} = \frac{1}{5}.$$

$$\textcircled{2} \quad \frac{1}{5} \cdot 500 = 100 \text{ people.}$$

$$\textcircled{3} \quad 72 \div 3.6 = 20\%.$$

A fifth ✓

ANSWER

100 people, or 20%

EXAMPLE 3 School A has 200 pupils and school B has 500. Football is 90° in both charts. Compare them.

$\textcircled{1}$ The shares are equal — a quarter each.

$\textcircled{2}$ 50 pupils in A, 125 in B.

$\textcircled{3}$ Same proportion, very different numbers.
The angle alone cannot compare schools.

ANSWER

Equal shares; 50 against 125 pupils

03 Interactive model

Show two charts side by side with the totals covered, ask which school has more footballers, then uncover the totals. The reversal is the point of the lesson.

Reading the chart

A 120° sector of 36 pupils is:

A 54° sector of 60 pupils is:

A 90° sector of 200 people is:

A 144° sector as a percentage is:

14.4%

40%

44%

60%

A 36° sector as a percentage is:

3.6%

10%

36%

20%

Without the total, a chart gives:

the numbers

the shares only

the mean

nothing at all

90° of 400 against 120° of 180:

100 against 60

60 against 100

equal

cannot tell

A pie chart cannot show:

the largest group

a share

the change over time

a comparison of two sectors

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
To interpret	Talqin qilmoq	Интерпретировать
Sector angle	Sektor burchagi	Угол сектора
Proportion	Ulush	Доля
Total surveyed	So'ralganlar soni	Число опрошенных
To compare	Taqqoslamog	Сравнивать
Conclusion	Xulosa	Вывод
Misleading	Chalg'ituvchi	Вводящий в заблуждение
Estimate	Taxminiy baho	Оценка

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

To turn a sector into a frequency you need:

the angle only

the angle and the total

the total only

a protractor

A 180° sector is:

a quarter

a half

a third

everything

To turn an angle into a percentage divide by:

3.6

36

360

100

Two charts with equal sectors represent:

equal numbers

equal shares

nothing comparable

equal totals

A larger sector always means:

more people

a larger share

a larger total

a better result

A good interpretation ends with:

a bigger chart

an honest limit

the mean

the angles

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. 36 pupils, a sector of 120°

ANSWER 12 pupils

2. 36 pupils, a sector of 90°

ANSWER 9 pupils

3. 60 pupils, a sector of 54°

ANSWER 9 pupils

4. 60 pupils, a sector of 36°

ANSWER 6 pupils

5. A sector of 180° as a fraction

ANSWER $\frac{1}{2}$

6. A sector of 90° as a percentage

ANSWER 25%

7. A sector of 36° as a percentage

ANSWER 10%

Medium · the standard question

7 PROBLEMS

1. 200 people, a sector of 90°

ANSWER 50 people

2. 500 people, a sector of 72°

ANSWER 100 people

3. 240 people, a sector of 45°

ANSWER 30 people

4. A sector of 144° as a percentage

ANSWER 40%

5. A sector of 54° as a percentage

ANSWER 15%

6. 90 people, a sector of 100°

ANSWER 25 people

7. 720 people, a sector of 30°

ANSWER 60 people

Hard · stretch and reason

7 PROBLEMS

1. A sector of 60° as a percentage, to 1 d.p.

ANSWER 16.7%

2. Football at 90° in schools of 200 and 500

ANSWER 50 and 125 pupils

3. 120° of 180 against 90° of 400

ANSWER 60 against 100

4. Which of those two has more footballers?

ANSWER The second, despite the smaller sector

5. What can a pie chart never give without more information?

ANSWER The number of people surveyed

6. A sector estimated at 100° out of 90 pupils

ANSWER 25 pupils

7. Two pie charts can be compared by number only when

ANSWER Both totals are known

07 Homework

Homework — 5 tasks

Write the total beside every chart you read; without it no frequency can be found.

1. A pie chart of 72 pupils has a 100° sector. How many pupils does it represent?
2. The same chart has a 45° sector. How many pupils is that?
3. Express 144° and 54° as percentages of the whole.
4. In school A (300 pupils) football is 120° ; in school B (600) it is 90° . Which school has more footballers?

5. Write two questions that a pie chart on its own cannot answer.